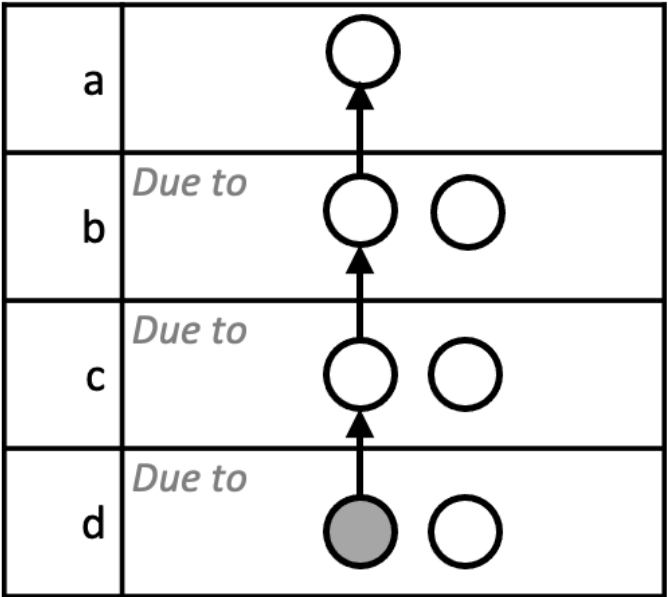
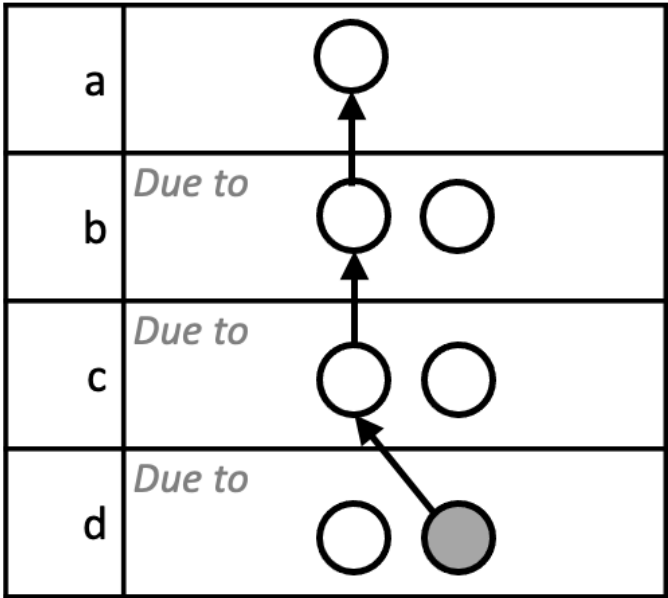
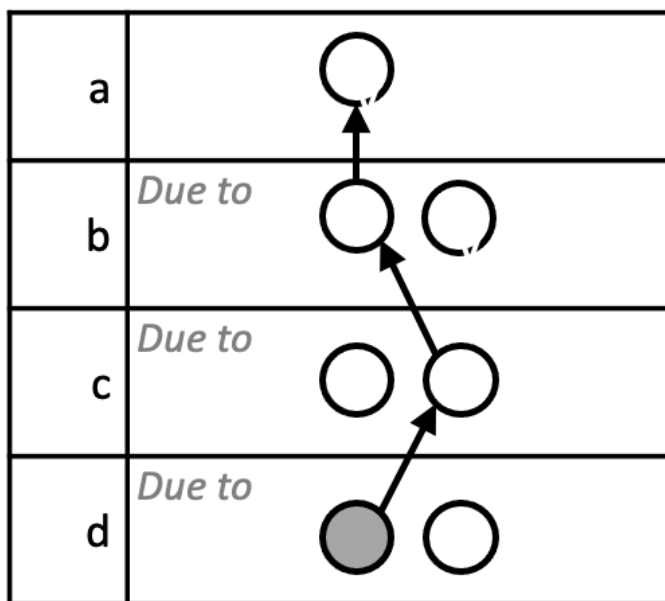




First Mentioned Sequence C



First Mentioned Sequence D



First Mentioned Sequence E

If there are still conditions reported on lower lines in Part 1, repeat the procedure for as long as a new tentative starting point can be identified. When no condition can be found that could cause the tentative starting point, the last identified tentative starting point is also the starting point of the first-mentioned sequence.

Example 1

- 1 (a) Pneumonia
due to
- (b) Hip fracture and heart failure
due to
- (c) Tripped on carpet, coronary atherosclerosis
due to
- (d)
- 2

Pneumonia can be due to hip fracture, and therefore hip fracture is the tentative starting point. Hip fracture can be due to tripping, which is the new tentative starting point. Since there are no causes reported below line 1(c), tripping on carpet is the starting point of the first-mentioned sequence.

Example 2

- 1 (a) Pneumonia
due to
 - (b) Heart failure and hip fracture
due to
 - (c) Coronary atherosclerosis and tripped on carpet
due to
 - (d)
- 2

Pneumonia can be due to heart failure, and therefore heart failure is the tentative starting point. Heart failure can be due to coronary atherosclerosis, which is the new tentative starting point. Since there are no causes reported below line 1(c), coronary atherosclerosis is the starting point of the first-mentioned sequence.

Example 3

- 1 (a) Pneumonia
due to
 - (b) Hip fracture and heart failure
due to
 - (c) Coronary atherosclerosis and tripped on carpet
due to
 - (d)
- 2

Pneumonia can be due to hip fracture, and therefore hip fracture is the tentative starting point. However, hip fracture cannot be due to coronary atherosclerosis, but hip fracture can be due to tripping, which is the new tentative starting point. Since there are no causes reported below line 1(c), tripped on carpet is the starting point of the first-mentioned sequence.

2.16.6 Underlying cause of death (UCOD)

The underlying cause of death (UCOD), as defined in Section [2.15.1](#), is the condition selected for single-cause tabulation of mortality statistics.

A condition that is provisionally considered as the underlying cause of death when applying the instructions step by step is referred to as a 'tentative underlying cause of death (TUC)' and may change several times as the instructions are applied to the death certificate.

Example 1

- 1 (a) Myocardial infarction
due to
- (b) Coronary atherosclerosis
due to
- (c) Generalised atherosclerosis
due to
- (d)
- 2

Generalised atherosclerosis started the sequence of events leading to death, so it is the starting point. There are special modification instructions relating to atherosclerosis and coronary heart disease in the ICD, and, in the next step, coronary atherosclerosis is selected as the tentative underlying cause of death. But there are further instructions on coronary atherosclerosis and myocardial infarction, and in the final step, myocardial infarction is selected as the tentative underlying cause, and is the underlying cause of death in this case.

2.16.7 Priority underlying condition

Some mortality coding instructions (e.g. Steps SP6, M1) refer to the 'priority underlying condition'. It is a concept to set a priority order giving precedence to the underlying condition, when specific requirements in each instruction apply to several conditions in the death certificate.

To identify the priority underlying condition, start from the first condition reported on the lowest used line of Part 1. If there are several conditions reported, search from the lowest used line, and the next line above in turn, and from left to right for each line. If you cannot find the priority underlying condition in Part 1, then search Part 2, again from left to right.

1	a		⑦	⑧	
	b	<i>Due to</i>	⑤	⑥	
	c	<i>Due to</i>	③	④	
	d	<i>Due to</i>	①	②	
2			⑨	⑩	

Priority Underlying Condition

2.16.8 Modification

Special coding instructions on specific sequences and ICD categories may have the effect that a condition other than the starting point is selected as the underlying cause of death for use in the statistics. In such cases, the code for underlying cause often expresses a combination of the starting point with another reported condition, or a complication or consequence of the starting point that is of particular importance to public health. The procedure by which the ICD code for the starting point is replaced by another code is called modification. Instructions on how to apply these special instructions to identify the underlying cause of death is given in Section [2.18 Check for modifications of the starting point \(Steps M1 to M4\)](#).

Example 1

- 1 (a) Heart disease
due to
 - (b) Generalised atherosclerosis
due to
 - (c)
due to
 - (d)
- 2

Generalised atherosclerosis started the sequence of events leading to death, so it is the starting point. However, according to a special instruction on generalised atherosclerosis, generalised atherosclerosis leading to heart disease is assigned to atherosclerotic heart disease in mortality statistics. Because of this modification, atherosclerotic heart disease is the underlying cause of death.

2.17 Coding instructions for mortality

When coding causes of death, first assign ICD codes to all the conditions mentioned on the death certificate. Many coding instructions are based on specific ICD codes, and to determine whether or not any of the instructions apply, all conditions on the certificate must be coded. Other conditions reported on the certificate may affect the coding any condition. Assigning codes for all reported conditions and applying any effect one code has on another is called multiple cause coding.

2.17.1 Basic coding and multiple cause coding guidelines

To start coding, refer to basic coding guidelines given in sections in the beginning of Part 2 (for example Sections [2.1](#), [2.6](#), and [2.10.1](#)). When multiple causes are reported also refer to Section [2.21 Coding instructions for mortality: multiple cause coding and other specific instructions](#). Multiple cause coding permits in-depth analysis of causes of death, for example of serious but avoidable complications of certain underlying causes, and the impact of coexisting conditions on the outcome of a disease process. Therefore, in mortality coding, both underlying cause and multiple causes should be recorded. Complete multiple cause coding is essential for a correct application of the ICD instructions for selection and modification of the underlying cause of death.

To ensure consistent reporting of emerging conditions of international public health importance, WHO may sometimes publish special instructions for coding of named conditions, or selection of the underlying cause of death where named conditions are mentioned on the Medical Certificate Cause of Death (MCCD). This will usually involve codes in Chapter 25: Codes for special purposes - International provisional assignment of new diseases of uncertain aetiology and emergency use. Such special instructions should be applied in preference to any instructions in the Reference Guide that would otherwise apply.

Once ICD codes are assigned to each disease or condition on the certificate, apply the instructions to select the underlying cause of death. Note that the full instructions include, in some cases, supplementary tables which are contained in the online Mortality Rules Knowledgebase as explained in Section [2.20](#)

2.17.2 Selecting the underlying cause of death

For most death certificates, selecting the underlying cause of death is a straightforward procedure. There are, however, many cases where the underlying cause is not immediately obvious. To ensure that both straightforward and complex cases are coded according to the ICD mortality rules, it is important to follow the coding instructions carefully, step by step. Otherwise, the resulting mortality statistics will not be internationally comparable, which seriously reduces the value of the data for public health purposes.

Selecting the underlying cause of death involves two separate steps. The first step is to identify the starting point (Steps SP1 through SP8 below). The next step is to modify the starting point, if any of the modification instructions apply, to reflect further information provided on the death certificate (Steps M1 through M4 below). See Sections [2.18](#) - [2.19](#) for specific instructions. In addition, Mortality Annex [ERROR](#) includes a workflow diagram to illustrate the coding instructions for the selection of the underlying cause of death. This is intended as a supplement to help coders follow the coding instructions.

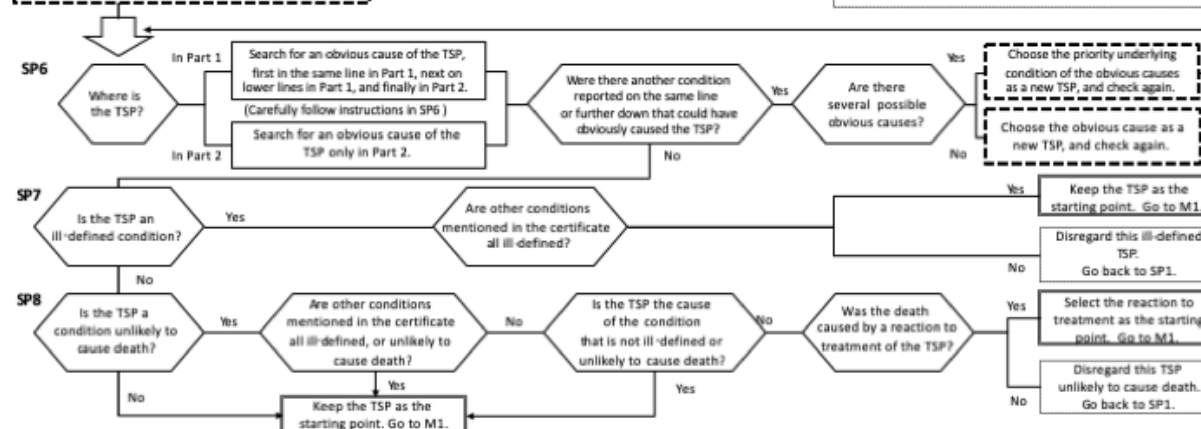
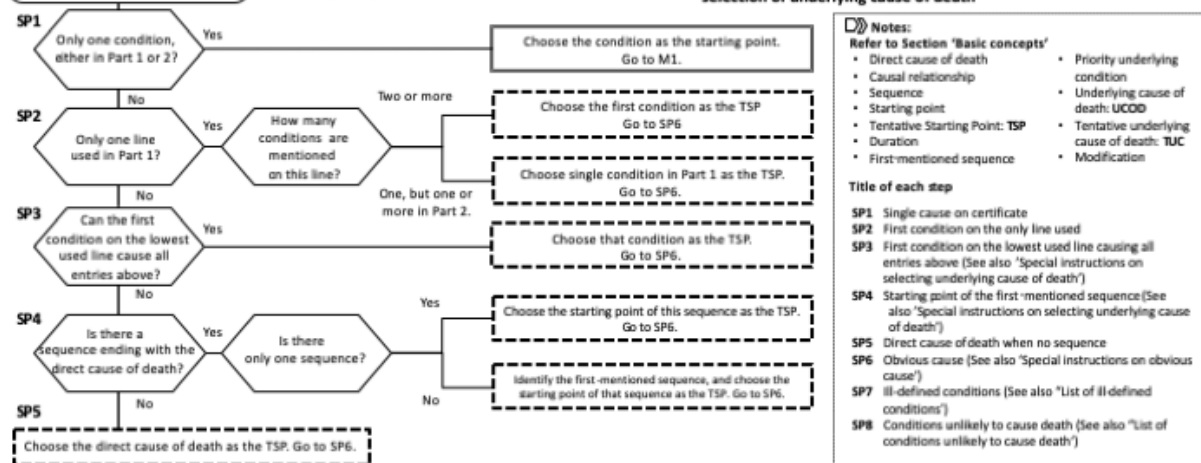
Note that the purpose of the selection procedure is to produce the most useful mortality statistics possible. Thus, the following instructions may reflect the importance of some conditions for public health rather than what is correct from a purely medical point of view. The following instructions always apply, whether they might be considered medically correct or not.

In the coding examples that follow, the 'due to' statement between the lines in Part 1 is not included. But it is important to bear in mind that anything reported on an upper line in Part 1 is meant to be due to what is reported on the line below.

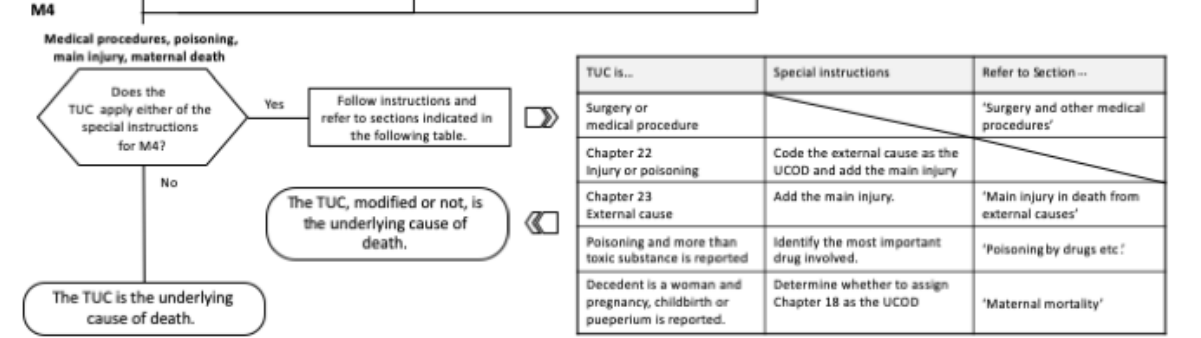
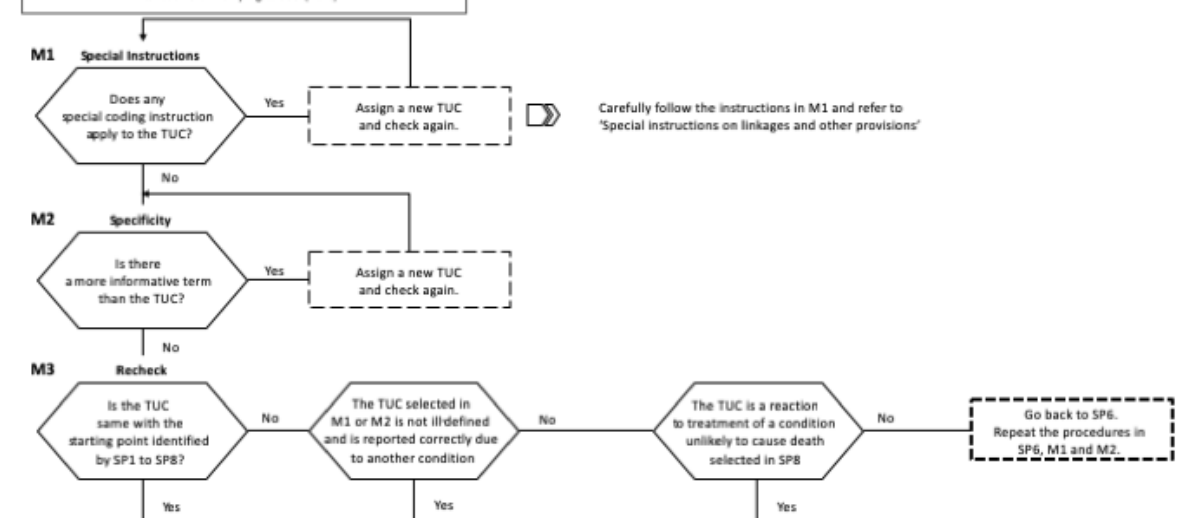
First assign ICD codes to all conditions mentioned on the death certificate

see Section 'Coding instructions for mortality: multiple causes'

Workflow diagram of the application of the coding instruction for the selection of underlying cause of death



Starting point identified by SP1 to SP8 is now considered the tentative underlying cause (TUC).



Workflow

Workflow diagram of steps SP1 to SP8, and to Steps M1 to M4 for mortality coding.

2.17.3 Find the starting point (Steps SP1 to SP8)

To identify the starting point, follow the eight steps specified in this section. The steps are named SP1 to SP8 (Starting point rule 1 to Starting point rule 8). Each step contains one selection rule. At each step, there is a description of the selection rule itself and an instruction on what to do next.

2.17.4 Step SP1 – Single cause on certificate

If there is only one condition reported on the certificate, in either Part 1 or Part 2, this is the starting point. Next verify whether either step M1 or M4 apply, go to Section [2.18 Check for modifications of the starting point \(Steps M1 to M4\)](#).

If there are two or more conditions on the certificate, go to Step SP2.

2.17.5 Step SP2 – First condition on the only line used

If the certifier has used only one line in Part 1 and:

- has reported only one condition on this line, but has reported one or more conditions in Part 2, then the single condition in Part 1 is the tentative starting point. Next, go to Step SP6.
- has reported two or more conditions on this line, then the first condition is the tentative starting point. This applies whether or not one or more conditions are reported in Part 2. Next, go to Step SP6.

If the certifier has used more than one line in Part 1, go to Step SP3.

If the certifier has used only Part 2, but has reported two or more conditions there, then the first condition is the tentative starting point.

Example 1

- 1 (a) Myocardial infarction and diabetes mellitus
- (b)
- (c)
- (d)
- 2

Myocardial infarction is mentioned first on the certificate and is the tentative starting point. Next, go to Step SP6, to check whether further selection and modification rules apply.

Example 2

- 1 (a) Myocardial infarction
(b)
(c)
(d)
- 2 Diabetes mellitus

Myocardial infarction is mentioned first on the certificate and is the tentative starting point. Next, go to Step SP6, to check whether further selection and modification rules apply.

2.17.6 Step SP3 – First condition on the lowest used line causing all entries above

If there are conditions reported on more than one line in Part 1, check if each of the conditions reported on the line(s) above the lowest used line in Part 1 can be caused by the first condition on the lowest used line.

- If yes, then this condition is the tentative starting point. Next, go to Step SP6.
- If not, go to Step SP4.

To assess causal relationship, refer to Section [2.16.2 Causal relationship and sequence](#), and to Section [2.19.1 Special instructions on accepted and rejected sequences \(Steps SP3 and SP4\)](#).

Example 1

- 1 (a) Bronchopneumonia
(b) Hemiplegia
(c) Cerebral infarction
(d)
- 2

Both bronchopneumonia and hemiplegia can be caused by cerebral infarction. This means that cerebral infarction is the tentative starting point.

Example 2

- 1 (a) Liver metastases 2 months
(b) Bronchopneumonia 4 days
(c) Stomach cancer 6 months
(d)
- 2

Both liver metastases and bronchopneumonia can be caused by stomach cancer. This means that stomach cancer is the tentative starting point, even though bronchopneumonia cannot cause liver metastases and the bronchopneumonia has a shorter duration than the liver metastases.

Example 3

- 1 (a) Liver metastases
(b) Bronchopneumonia
(c) Stomach cancer and cerebral infarction
(d)
- 2

Both liver metastases and bronchopneumonia can be caused by stomach cancer, which is the first condition mentioned on the lowest used line in Part 1. This means that stomach cancer is the tentative starting point.

Example 4

- 1 (a) Liver metastases
(b) Bronchopneumonia and stomach cancer
(c)
(d)
- 2

Liver metastases cannot be due to bronchopneumonia. This means that no tentative starting point can be identified at Step SP3. Therefore, go to Step SP4.

2.17.7 Step SP4 – Starting point of the first-mentioned sequence

The first-mentioned sequence is always found in Part 1 (see Section [2.16.5](#)).

- If one or more sequences ending with the terminal cause of death are reported in Part 1, identify the first-mentioned sequence (see Section [2.16.5](#)). The starting point of this sequence is the tentative starting point. Next go to Step SP6.

If no sequence ending with the terminal cause of death is reported in Part 1, go to Step SP5.

To assess causal relationship, refer to Section [2.16.2 Causal relationship and sequence](#), and to Section [2.19.1 Special instructions on accepted and rejected sequences \(Steps SP3 and SP4\)](#).

Example 1

- 1 (a) Liver metastases
(b) Bronchopneumonia and stomach cancer
(c)
(d)
- 2

Bronchopneumonia cannot cause liver metastases (Step SP3 does not apply), but liver metastases can be due to stomach cancer. This is the first-mentioned sequence ending with the terminal cause of death, so stomach cancer is the tentative starting point.

Example 2

- 1 (a) Bronchopneumonia
- (b) Cerebral infarction and liver metastases
- (c) Atherosclerosis and stomach cancer
- (d)

2

Atherosclerosis cannot cause liver metastases (Step SP3 does not apply). There are three acceptable sequences on the certificate: (1) bronchopneumonia caused by cerebral infarction, in its turn caused by atherosclerosis; (2) bronchopneumonia caused by cerebral infarction, in its turn caused by stomach cancer; and (3) bronchopneumonia caused by liver metastases, in its turn caused by stomach cancer. The first-mentioned sequence is bronchopneumonia caused by cerebral infarction, in its turn caused by atherosclerosis. Consequently, atherosclerosis is the tentative starting point.

2.17.8 Step SP5 – Terminal cause of death when no sequence

If no sequence ending with the terminal cause of death is reported in Part 1, the terminal cause of death is also the tentative starting point. Next, go to Step SP6.

Example 1

- 1 (a) Liver metastases
- (b) Cerebral infarction
- (c) Atherosclerosis
- (d)

2 Stomach cancer

Atherosclerosis cannot cause liver metastases (Step SP3 does not apply). Cerebral infarction cannot cause liver metastases (Step SP4 does not apply). No sequence ending with the terminal cause of death is reported in Part 1, so the terminal cause of death – liver metastases – is the tentative starting point.

2.17.9 Step SP6 – Obvious cause

If the tentative starting point selected in Steps SP1 to SP5 was obviously caused by another condition on the certificate, select the obvious cause as the new tentative starting point. Conditions that are considered to have an 'obvious' causal relationship are specified in Section [2.19.2 Special instructions on obvious cause \(Step SP6\)](#). To identify which Part of the certificate you should search for, apply following rules:

- If the tentative starting point is in Part 1, the obvious cause must be either on the same or lower line, in Part 1, or in Part 2. Do not look for obvious causes on lines above the tentative starting point.
- If the tentative starting point is in Part 2, the obvious cause must also be in Part 2. Do not look for obvious causes in Part 1.

Next, reapply Step SP6 to the new tentative starting point. Continue looking for a new tentative starting point until you find a tentative starting point that is not obviously caused by a condition reported on the same line or further down on the certificate. Then go to Step SP7.

If there is no condition reported on the certificate that obviously caused the tentative starting point selected in Steps SP2 to SP5, go to Step SP7.

An obvious causal relationship is a type of causal relationship (see Section [2.19.1](#)) and a rejected sequence in Section [2.19.2] should also be rejected in considering an 'obvious' relationship between conditions. However, in considering accepting sequences, the word 'obviously' is important, and there must be no doubt about the relationship between the conditions. It is not sufficient that the sequence would have been accepted if the tentative starting point had been reported as due to the other condition. In applying Step SP6, always refer to Section [2.19.2](#).

Do not apply Step SP6 if the tentative starting point has a longer duration than the obvious cause.

If more than one obvious cause of the tentative starting point is reported, select the priority underlying condition (see Section [2.16.7](#)).

Example 1

- 1 (a) Sepsis
(b) Peritonitis
(c)
(d)
- 2 Appendicitis with rupture

Sepsis can be caused by peritonitis, and peritonitis is the tentative starting point (Step SP3). Appendicitis with rupture is an obvious cause of peritonitis, and appendicitis with rupture is the new tentative starting point.

Example 2

- 1 (a) Liver metastases
(b) Cerebral infarction
(c)
(d)
- 2 Stomach cancer

Cerebral infarction cannot cause liver metastases, and liver metastases is the tentative starting point (Step SP5). Stomach cancer is an obvious cause of liver metastases, and stomach cancer is the new tentative starting point.

Example 3

- 1 (a) Sepsis
(b) Peritonitis, mesenteric embolism
(c)
(d)
- 2

Sepsis can be caused by peritonitis, and peritonitis is the tentative starting point (Step SP3). Mesenteric embolism is an obvious cause of peritonitis, and mesenteric embolism is the new tentative starting point.

Example 4

- 1 (a) Sepsis
(b) Peritonitis
(c)
(d)
- 2 Mesenteric embolism, rupture appendicitis

Sepsis can be caused by peritonitis, and peritonitis is the tentative starting point (Step SP3). Next, both mesenteric embolism and ruptured appendicitis are obvious causes of peritonitis. Because mesenteric embolism is mentioned first and is the priority underlying condition, it is the new tentative starting point.

Example 5

- 1 (a) Sepsis
(b) Peritonitis
(c)
(d)
- 2 Necrosis of intestine, mesenteric infarction

Sepsis can be caused by peritonitis, and peritonitis is the tentative starting point (Step SP3). Necrosis of intestine is an obvious cause of peritonitis, so necrosis of intestine is the new tentative starting point. Next, mesenteric infarction is an obvious cause of necrosis of intestine, and mesenteric infarction is the final starting point.

2.17.10 Step SP7 – Ill-defined conditions

If the tentative starting point selected in Steps SP1 to SP6 is in the [List of ill-defined conditions](#), and:

- If there is at least one condition that is not ill-defined, then disregard the ill-defined condition. Go to Step SP1 and select another starting point, as if the ill-defined condition had not been mentioned on the certificate.
- If all other conditions reported on the certificate, are ill- defined, go to Step M1.

If the tentative starting point is not ill-defined, go to Step SP8.

Note that the following are not considered ill-defined:

[3.14.6 List of ill-defined conditions.](#)

Example 1

- 1 (a) Respiratory failure
(b)
(c)
(d)
- 2 Mesenteric embolism

Respiratory failure is the only condition mentioned in Part 1 and it is the tentative starting point according to Steps SP2. Respiratory failure is in the table of ill-defined conditions, and there is a condition not ill-defined, Mesenteric embolism, so disregard respiratory failure and restart the selection procedure. Mesenteric embolism is the new starting point according to Step SP1.

2.17.11 Step SP8 – Conditions unlikely to cause death

If the tentative starting point selected in Steps SP1 to SP7 is in the [List of conditions unlikely to cause death](#) (see Mortality Annex [3.14.10](#), and:

- If all other conditions reported on the certificate are also unlikely to cause death or ill-defined, then keep this condition 'unlikely to cause death' as the starting point. Next, go to Step M1.
- If this condition was the cause of another condition that is not 'unlikely to cause death' and that is not ill-defined, then keep this condition unlikely to cause death as the starting point. Next, go to Step M1.
- If the death was caused by a reaction to treatment of the condition unlikely to cause death, select the reaction to treatment as the starting point. Next, go to Step M1
- If none of these three apply, and there is at least one condition that is not 'unlikely to cause death' and not 'ill-defined', then disregard the condition unlikely to cause death. Go to Step SP1 and select another starting point, as if the condition unlikely to cause death had not been mentioned on the certificate.

If the tentative starting point is not in the 'List of conditions unlikely to cause death', keep that condition as the starting point and go to Step M1.

If the certificate mentions more than one treatment for the condition unlikely to cause death, select the first-mentioned treatment.

Example 1

- 1 (a) Hearing loss
(b)
(c)
(d)
- 2 Ischaemic heart disease

Hearing loss is the tentative starting point (Step SP2), but it is in the 'List of conditions considered unlikely to cause death'. There is another condition on the certificate, ischaemic heart disease, which is not in the 'List of conditions considered unlikely to cause death'. Disregard hearing loss and restart the selection procedure from Step SP1. Ischaemic heart disease is the new starting point (Step SP1).

Example 2

- 1 (a) Liver failure
(b) Excessive use of paracetamol
(c) Migraine-type headache
(d)
- 2

Migraine type headache is the tentative starting point (Step SP3), but it is in the 'List of conditions considered unlikely to cause death'. Migraine type headache was treated with paracetamol and there was a reaction to the treatment, liver failure. Select the reaction to the treatment, liver failure, as the starting point.

Example 3

- 1 (a) Sepsis
(b) Submandibular abscess
(c) Caries
(d)
- 2

Caries is the tentative starting point (Step SP3), but it is in the 'List of conditions considered unlikely to cause death'. In this case caries caused submandibular abscess, a condition that is not unlikely to cause death and that is not ill-defined. Because of that, keep caries as the starting point.

Example 4

- 1 (a) Headache
(b) Caries
(c)
(d)
- 2 Ischaemic heart disease

Caries is the tentative starting point (Step SP3), but it is in the 'List of conditions considered unlikely to cause death'. In this case caries caused headache, a condition that is in the list of ill-defined conditions. Disregard both caries (Step SP8) and headache (Step SP7) and restart the selection procedure from Step SP1. Ischaemic heart disease is the new starting point (Step SP1).

2.18 Check for modifications of the starting point (Steps M1 to M4)

The starting point identified using Steps SP1 to SP8 is now considered the tentative underlying cause. There may be special coding instructions on this tentative underlying cause, or other reasons to modify the tentative underlying cause. Check whether the tentative underlying cause should be modified by applying the modification rules described in steps M1 to M3 (Modification rule 1 to Modification rule 3). Each step contains one modification rule. At each step, there is a description of the modification rule itself and what to do next. There are also bullet points with more detailed instructions and explanations.

2.18.1 Step M1 – Special instructions

If the tentative underlying cause (TUC) selected in Steps SP1 to SP8 applies to a special instruction listed in Section [2.19.3 Special instructions on linkages and other provisions \(Step M1\)](#), assign a new tentative underlying cause according to the instruction.

Next, reapply Step M1 to the new tentative underlying cause. Repeat until you have found a tentative underlying cause that is not affected by any further special coding instruction. Next, go to Step M2.

If the tentative underlying cause does not apply to instructions in Section [2.19.3](#), go to Step M2.

If more than one instruction in Section [2.19.3](#) applies to the tentative underlying cause, select the instruction relating to the priority underlying condition (see Section [2.16.7](#)).

Note that there are two types of combinations of codes, 'with mention of' and 'when reported as a cause of'. Refer to Section [2.19.3](#) for details.

Sometimes the classification itself indicates a code for a combination of the tentative underlying cause with another cause mentioned on the certificate. Use the combination code unless an instruction on mortality coding in Section [2.19.3](#) indicates otherwise.

Examples of 'with mention of':

Example 1

- 1 (a) Myocardial infarction
- (b) Ischaemic heart disease
- (c)
- (d)

2

Ischaemic heart disease is the tentative starting point (Step SP3). There is a special instruction on ischaemic heart disease reported with mention of myocardial infarction, and, according to this instruction, myocardial infarction is the new tentative underlying cause.

Example 2

- 1 (a) Ischaemic heart disease
- (b) Atherosclerosis
- (c)
- (d)

2 Cerebral infarction

Atherosclerosis is the tentative starting point (Step SP3). There is a special instruction on 'atherosclerosis reported with ischaemic heart disease', and another one on 'atherosclerosis reported with cerebral infarction'. Ischaemic heart disease is the priority underlying condition, so apply the instruction on 'atherosclerosis reported with ischaemic heart disease' and select ischaemic heart disease as the new tentative underlying cause.

Example 3

- 1 (a) Cerebrovascular infarction
- (b) Atherosclerosis
- (c) Hypertension
- (d)

2 Myocardial infarction

Hypertension is the tentative starting point (Step SP3). There are special instructions on 'hypertension reported with mention of cerebrovascular infarction' and with myocardial infarction. Cerebrovascular infarction is the priority underlying condition, so apply the instruction on 'hypertension reported with mention of cerebrovascular infarction' and select cerebrovascular infarction as the new tentative underlying cause.

Example 4

- 1 (a) Ischaemic heart disease
(b) Atherosclerosis
(c)
(d)
- 2 Myocardial infarction

Atherosclerosis is the tentative starting point (Step SP3). There is a special instruction on 'atherosclerosis reported with mention of ischaemic heart disease', and another one on 'atherosclerosis reported with mention of myocardial infarction'. Ischaemic heart disease is the priority underlying condition, so apply the instruction on 'atherosclerosis reported with mention of ischaemic heart disease' and select ischaemic heart disease as the new starting point. Next, there is a special instruction on 'ischaemic heart disease reported with mention of myocardial infarction'. Apply this instruction and select myocardial infarction as the new tentative underlying cause.

Examples of 'when reported as the cause of':

Example 5

- 1 (a) Chronic kidney disease
(b) Atherosclerosis
(c)
(d)
- 2

Atherosclerosis is the tentative starting point (Step SP3). There is a special instruction on 'atherosclerosis reported as the cause of chronic kidney disease'. Apply this instruction and select hypertensive renal disease ([BA02](#)) as the new tentative underlying cause.

Example 6

- 1 (a) Atherosclerosis
(b)
(c)
(d)
- 2 Chronic kidney disease

Atherosclerosis is the tentative starting point (Step SP2). Although there is a special instruction on 'atherosclerosis reported as the cause of chronic kidney disease', this instruction does not apply here because chronic kidney disease is reported in Part 2 and not as caused by atherosclerosis. In this case, atherosclerosis remains the tentative starting point.

2.18.2 Step M2 – Specificity

If the tentative underlying cause describes a condition in general terms and another term that provides more precise information about the site or nature of the same condition is reported on the certificate, use the more informative term. This rule will often apply when the general term becomes an adjective, qualifying the more precise term.

If the tentative underlying cause is an infectious disease reported as the cause of a specific manifestation, there may be a provision in the classification to combine the two codes into

one code providing more precise information about the site or nature of the tentative starting point. This code is the new tentative underlying cause of death.

Next, reapply Step M2 to the new tentative underlying cause. Repeat until you have found a tentative underlying cause that cannot be specified further.

- If there is no term that further specifies the tentative underlying cause (TUC), go to Step M3.
- If there are several other expressions that provide more precise information on the tentative underlying cause, select the priority underlying condition (see Section [2.16.7](#)).
- If the TUC is an unspecified condition and this condition appears with more specificity somewhere on the death certificate, this last more specific condition should be considered as the new TUC.

Example 1

- 1 (a) Meningitis [1D01.Z](#)
(b) Tuberculosis [1B1Z](#)
(c)
(d)
- 2

Tuberculosis is the tentative starting point (Step SP3). A term providing more precise information about the site or nature of this condition is reported on the certificate: meningitis. There is a provision in the classification to combine the two codes, and according to instruction of step M2, tuberculous meningitis ([1B11.0](#)) is a new tentative underlying cause.

Example 2

- 1 (a) Syphilis, unspecified [1A6Z](#)
(b)
(c)
(d)
- 2 Primary genital syphilis [1A61.0](#)

Syphilis, is the tentative starting point (Step SP2). A more specific condition for syphilis appears elsewhere on the death certificate: primary genital syphilis, and according to instruction of step M2, primary genital syphilis is a new tentative underlying cause.

Example 3

- 1 (a) Acute gastrointestinal bleeding [ME24.90](#)
(b) Malignant neoplasms of colon, unspecified [2B90.Z](#)
(c)
(d)
- 2 Malignant neoplasm of transverse colon [2B90.2](#)

Malignant neoplasms of colon, is the tentative starting point (Step SP3). A more specific condition appears elsewhere on the death certificate: malignant neoplasm of transverse colon, and according to instruction of step M2, malignant neoplasm of transverse colon [2B90.2](#) is a new tentative underlying cause.

Regarding coding, this specificity can occur even between parents:

Example 4

- 1 (a) Sepsis without septic shock [1G40](#)
(b) Diabetes mellitus [5A14](#)
(c)
(d)
- 2 Type 1 diabetes mellitus [5A10](#)

Diabetes mellitus, type unspecified, is the tentative starting point (Step SP3). A more specific condition appears elsewhere on the death certificate: Type 1 diabetes mellitus, and according to instruction of step M2, Type 1 diabetes mellitus [5A10](#) is a new tentative underlying cause.

Example 5

- 1 (a) Stroke [8B20](#)
(b)
(c)
(d)
- 2 Intracerebral Haemorrhage [8B00](#)

Stroke, not known if ischaemic or haemorrhagic, is the tentative starting point (Step SP2). A term providing more precise information about the nature of this condition is reported on the certificate: intracerebral haemorrhage. According to instruction of step M2, intracerebral haemorrhage [8B00](#) is a new tentative underlying cause.

Example 6

- 1 (a) Acute pancreatitis [DC31.Z](#)
(b) Cytomegaloviral disease [1D82.Z](#)
(c)
(d)
- 2

Cytomegaloviral disease, unspecified, is the tentative starting point (Step SP3). The manifestation is described as acute pancreatitis, unspecified, and the two terms combine into cytomegaloviral pancreatitis ([1D82.1](#)), which is the new tentative underlying cause.

Special instructions on Chapter 21 Symptoms, signs or clinical findings, not elsewhere classified

TUC is:

[MH10](#) Brain death

[MH14](#) Other ill-defined or unspecified causes of mortality

with mention of:

Ill-defined conditions excluding [MH10](#)

Ill-defined conditions excluding [MH14](#)

code to:

Ill-defined conditions excluding [MH10](#)

Ill-defined conditions excluding [MH14](#)

2.18.3 Step M3 – Recheck Steps SP6, M1 and M2

If, at this point, the tentative underlying cause is not the same as the starting point you selected in Steps SP1 to SP8, then go back to Step SP6. Repeat the procedures described in Steps SP6, M1 and M2.

If the tentative underlying cause is the same with the starting point selected in Step SP1 to SP8, go to Step M4.

- Do not go back to Step SP6 if the cause selected in Step M1 or M2 is correctly reported as due to another condition, except when this condition is ill-defined.
- Also, do not go back to Step SP6 if the tentative underlying cause is a reaction to treatment of a condition unlikely to cause death, as selected in Step SP8.

Example 1

- 1 (a) Sepsis
(b) Arterial disease, arterial embolism of left leg
(c)
(d)
- 2 Colon cancer

Arterial disease is the tentative starting point according to Step SP3. Arterial embolism of left leg, reported as the second condition on line 1(b), is a specific type of arterial disease. Therefore, select arterial embolism of left leg as the tentative underlying cause in Step M2. Reapply Step SP6, because the tentative starting point is not the same as the one selected in Steps SP1 to SP8. But colon cancer is an obvious cause of arterial embolism, and colon cancer is the new starting point. No further modifications apply. Code colon cancer ([2B90.Z](#)), Malignant neoplasms of colon, unspecified) as the underlying cause of death.

Example 1

- 1 (a) Sepsis
(b) Arterial disease, arterial embolism of left leg
(c) Atherosclerosis
(d)
- 2 Colon cancer

Atherosclerosis is the tentative starting point (Step SP3). There is a special instruction on 'atherosclerosis reported as the cause of arterial disease', and, according to this instruction, arterial disease is the new starting point according to Step M1. Arterial embolism of left leg, reported as the second condition on line 1(b), is a more specific description of the type of arterial disease and is selected as the tentative starting point in Step M2. Do not reapply Step SP6, because arterial embolism of left leg is reported as due to atherosclerosis, and this is a correct causal relationship. No further modifications apply. Code 'arterial embolism of left leg' as the underlying cause of death.

2.18.4 Step M4 - Instructions on medical procedures, main injury, poisoning, and maternal deaths

Finally, apply the following instructions to the tentative underlying cause selected by applying Steps SP1 to SP8 and Steps M1 to M3.

If the tentative underlying cause is:

- Surgery, another type of medical procedure, a complication or postprocedural condition, apply the instructions in Section [2.19.4 Special instructions on surgery and other medical procedures \(Step M4\)](#).
- In Chapter 22 'Injury, poisoning or certain other consequences of external causes', first code the external cause of the injury or poisoning as the underlying cause of death. Also add the main injury to the cluster by following instructions in Section [2.19.5 Special instructions on main injury in deaths from external causes \(Step M4\)](#).
- In Chapter 23 'External causes of morbidity and mortality' also add the main injury to the cluster by following instructions in Section [2.19.5 Special instructions on main injury in deaths from external causes \(Step M4\)](#).
- Poisoning, use additional extension code from Section X (Extension Codes), if applicable, to identify the specific name of drug or toxic substance reported. If more than one drug or toxic substance is reported on the certificate, apply instructions in Section [2.19.6 Special instructions on poisoning by drugs, medications and biological substances \(Step M4\)](#), to identify the drug, medicament or substance most likely to have caused the death.

If the decedent is a woman, and pregnancy, childbirth or puerperium is reported on the certificate, determine whether to code the tentative underlying cause to Chapter 18 'Pregnancy, childbirth and the puerperium' according to the instructions in Section [2.19.7 Special instructions on maternal mortality \(Step M4\)](#).

When creating a cluster in Step M4, always put the code for the underlying cause of death at the beginning of the cluster.

After applying Step M4, the tentative underlying cause, modified or not, becomes the underlying cause of death.

If Step M4 does not apply, the tentative underlying cause becomes the underlying cause of death.

Note that other restrictions may apply, for example that the cause is limited to one of the sexes (see also Sections [2.21.7](#) and [3.14](#) (3.14.11 and 3.14.12)) or to a specific age range, or that the cause of death is improbable, considering the geographical setting. Therefore, always check whether any such restrictions apply to the underlying cause you selected.

2.19 Special instructions on selecting the underlying cause of death

The following sections are to be referred to in applying each instruction of Section [2.17.3](#) (Steps SP1 to SP8) and Section [2.18](#) (Steps M1 to M4).

2.19.1 Special instructions on accepted and rejected sequences (Steps SP3 and SP4)

This section lists sequences of causes of death that should be accepted or rejected when selecting the underlying cause of death. As described in Section [2.16.2 Causal relationship and sequence](#), these instructions are set with the aim of producing the most useful mortality statistics. Individual countries should not correct what is assumed to be an error, since changes at the national level will lead to data that are less comparable to data from other countries, and thus less useful for analysis.

A reported causal relationship not listed as rejected in this section should be accepted, as far as possible, because the certifier's opinion about the causes leading to death should not be disregarded lightly.

When applying Steps SP3 and SP4, reject the relationships listed in this section. Exceptions are listed as 'accept' in the table following each instruction.

Note that all information on causal relationship provided on the certificate should be considered. This applies also if the information appears in the 'wrong' place of the certificate. For example, if the sequence in Part 1 starts with a disease 'A', and information elsewhere on the certificate states that disease 'A' was due to a disease 'B', then consider 'B' as the tentative starting point. This also applies if disease 'A' and disease 'B' are coded to the same ICD-11 code and reported on separate lines of Part 1.

2.19.1.1 Conflicting durations

Do not accept a condition with a stated duration as due to a condition with a shorter duration.

Consequence	Caused by
A condition with a stated duration	Do not accept a condition with a shorter duration

When a duration of a condition is unknown or not stated assume the duration do not conflict, and do not reject a causal relationship by duration in such case.

2.19.1.2 Infectious diseases due to other conditions

Cholera and certain infectious diseases due to other conditions

Do not accept the following infectious and parasitic diseases as due to any other causes, **not even** human immunodeficiency virus (HIV) disease, malignant neoplasms or conditions impairing the immune system:

Condition	Code
1A00	Cholera
1A11	Botulism
1B20	Leprosy
1B50	Scarlet fever
1B91	Leptospirosis
1B93	Plague
1B94	Tularaemia
1B95	Brucellosis
1B97	Anthrax
1C11.1	Trench fever
1C12	Whooping cough
1C13	Tetanus
1C14	Obstetrical tetanus
1C15	Tetanus neonatorum
1C17	Diphtheria
1C1C	Meningococcal disease
1C22	Infections due to Chlamydia psittaci
1C23.Z	Trachoma
1C3Z	Rickettsioses
1C80-1C8Z	Viral infections of the central nervous system
1D20-1D2Z	Dengue
1D47	Yellow fever
1D40	Chikungunya virus disease
1D42	O'nyong-nyong fever
1D44	Rift Valley fever
1D46	West Nile virus infection
1D48	Zika virus disease
1D49	Crimean-Congo haemorrhagic fever
1D4A	Omsk haemorrhagic fever
1D4B	Kyasanur Forest disease
1D4C	Alkhurma haemorrhagic fever
1D40-1D4Z	Certain arthropod-borne viral fevers
1D60.0	Ebola disease
1D60.1	Marburg disease
1D61.0	Argentinian haemorrhagic fever
1D61.1	Bolivian haemorrhagic fever
1D61.2	Lassa fever

Condition	Code
1D62.0	Haemorrhagic fever with renal syndrome
1D65	Severe acute respiratory syndrome
1D80	Mumps
1D86	Viral haemorrhagic fever, not elsewhere classified
1E30-1E32	Influenza
1E50.1	Acute hepatitis B
1E50.2	Acute hepatitis C
1E51.0	Chronic hepatitis B
1E51.1	Chronic hepatitis C
1E51.2	Chronic hepatitis D
1E70	Smallpox
1E71	Mpox (Monkeypox)
1F02	Rubella
1F03	Measles
1F40-1F4Z	Malaria
1F51	African trypanosomiasis
1F53	Chagas disease
1F54	Leishmaniasis
8A45.01	Subacute sclerosing panencephalitis
8E02.0	Genetic Creutzfeldt-Jakob disease
R -	Other emerging diseases reportable to WHO (e.g. RA01 .- COVID-19)

Consequence condition **Causal condition**
Cholera etc., listed above **Do not accept** other causes

Typhoid and certain infectious disease due to other conditions

Do not accept the following infectious diseases as due to other causes, **except** HIV disease, malignant neoplasms and conditions impairing the immune system:

- [1A02](#) Intestinal infections due to Shigella
- [1A07](#) Typhoid fever
- [1A08](#) Paratyphoid fever
- [1A09](#) Infections due to other Salmonella
- [1B10-1B1Z](#) Tuberculosis

Consequence condition	Causal condition
Intestinal infections due to Shigella	Accept HIV disease, malignant neoplasms, and conditions impairing the immune system
Typhoid fever	Do not accept other causes
Paratyphoid fever	
Infections due to other Salmonella	
Tuberculosis	

HIV due to other conditions

Do not accept HIV disease (1C60 - 1C62) as due to other conditions, **except**:

- conditions necessitating blood transfusion, such as haemophilia, anaemia and major injuries
- invasive procedures, such as surgery
- drug abuse

Examples of such conditions are given in the Mortality Annex [3.14](#). Note that the list in Mortality Annex [3.14](#) is not complete and should be considered indicative.

Consequence condition	Causal condition
HIV	Accept - conditions necessitating blood transfusion, such as haemophilia, anaemia and major injuries - invasive procedures, such as surgery - drug abuse (for examples, refer to Mortality Annex 3.14.) Do not accept other causes

Infectious diseases not listed above due to other conditions

Infectious diseases not listed above are accepted to be caused by other conditions.

Consequence condition	Causal condition
Infectious diseases not listed above	Accept other causes

2.19.1.3 Malignant neoplasms due to other conditions

Do not accept a malignant neoplasm as due to any other cause, except :